Uchaaran

A little practice. A deeper connection.

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The problem: helping children learn Gurbani

Children need pronunciation support and a reason to practice. Conversations with Khalis Foundation focused us on engagement and understanding: help children return, recite with care, and explore what the words mean.

Two modes. One path toward mastery.

1. Practice at your pace

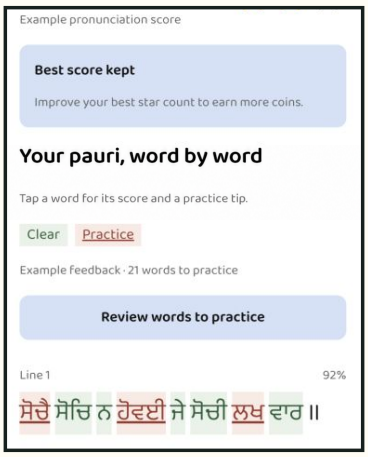
Listen and recall with Gurmukhi, Roman-letter Punjabi and English meaning. Revisit a line or word.

2. Take the pauri challenge

Recite a line and tap Done. Review word feedback. Earn stars and Nanak coins for character rewards.



Listen + recall



Review each word

What changed since midterm

Organizers asked for more attainable goals and simpler feedback for students. We added line-by-line scoring, short prompts, and an ML maintenance guide for Khalis Foundation.

Why a memory palace?

For grades 6 to 8, familiar scenes link each line to a place along 38 pauris. Sikh-inspired journeys invite recall and reflection. The ustaad guides pronunciation; rewards follow learning, not screen time.

Tools and implementation

Expo · React Native · TypeScript · Azure-hosted IndicConformer · BaniDB · reference audio. Codex and Paper supported development and design.

Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis Foundation can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

Watch demo ↗ | **Try the app ↗** | **View code ↗** | **Design process ↗**

Ready to help Khalis Foundation launch. An implementation PR is open in the main app repository for review and deployment. We can help Khalis Foundation set up infrastructure and support its technical owners.